

Paulo Eduardo dos Santos

SENIOR DATA ENGINEER | DATA PLATFORM ENGINEERING | REAL-TIME STREAMING, LAKEHOUSE & AWS | FINTECH & DIGITAL ASSETS

ATS Match: 90.0% — Data Platform Engineering Manager @ kraken

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PROFESSIONAL SUMMARY

Senior Data Engineer with 9 years building and operating large-scale **data platforms** for banks, payments, telecom, healthcare, and e-commerce organizations that process millions of daily transactions. Specialized in **AWS** data-lakes and lakehouse architecture (**Spark**, **EMR/EMR Serverless**, **Athena**, **Iceberg**, **Parquet**, **S3**, **Glue**, **Trino**), **dimensional modeling**, **data quality** best practices, and real-time / **streaming** data pipelines (**Kafka**, Kinesis, **CDC**, event-driven).

Hands-on in **Python**, **SQL**, **PySpark/Spark**, and **Scala**, with technical leadership and mentoring of engineering teams across multi-stakeholder delivery (product, finance, compliance, analytics). Proven results: **99.9% uptime SLA**, **60% storage reduction**, **50% faster processing**, **70% less manual operation**, and **40% maintenance reduction**. Builds platform capabilities that serve downstream analytics and **machine learning** consumers, and maintains public open-source **blockchain data** projects (Ethereum, Solana, DeFi) bridging enterprise data engineering with Web3/crypto and high-throughput financial data.

TECHNICAL SKILLS

Languages: Python (advanced), SQL (advanced), PySpark, Scala, Shell Script

AWS (cloud-native data platform): S3, EMR / EMR Serverless, Glue, Athena, Trino/Presto, Redshift, Lambda, Kinesis, Lake Formation, DMS, Step Functions, KMS, CloudWatch

Streaming & Real-Time: Apache Kafka, Kinesis, CDC (Change Data Capture), Debezium-style ingestion, Spark Streaming, event-driven architecture, Apache NiFi

Lakehouse & Formats: Apache Iceberg, Delta Lake, Apache Hudi, Parquet

Processing & Orchestration: Apache Spark, PySpark, SparkSQL, Apache Airflow, dbt, Talend, DataStage, SSIS, Informatica PowerCenter/BDM, Sqoop, Flume

Architecture & Modeling: Data Lake & Lakehouse (Medallion Bronze/Silver/Gold), Data Warehouse (Star/Snowflake schema), Dimensional Modeling, CDC, Event-Driven, Cloud Migration (on-prem → cloud), Big Data (Hadoop/Spark)

Governance & Data Quality: Data Quality frameworks, Data Governance, Data Catalog (AWS Glue, Unity Catalog), self-service tooling, PII/LGPD compliance, HashiCorp Vault, AWS KMS, Fernet encryption

Other Cloud: Azure (Data Lake, Databricks, Data Factory, Synapse, Unity Catalog, Fabric) · GCP (BigQuery, Dataproc)

Databases: Snowflake, SQL Server, Oracle, Teradata, PostgreSQL, MySQL, Redshift, MongoDB, Cassandra, HBase, DynamoDB

DevOps & IaC: Docker, Kubernetes, Terraform, Ansible, CI/CD, Git

Visualization: Power BI, QlikSense, Tableau, Spotfire, MicroStrategy, OBIEE, Grafana

Adjacent (working knowledge): Machine Learning fundamentals, AI/automation in data workflows (AI-assisted pipeline tooling)

Methodologies & Leadership: Agile for Analytics (Scrum, SAFe), technical leadership and mentoring of engineers, requirements gathering for analytics projects

KEY ACHIEVEMENTS

- Sustained a **99.9% uptime SLA** for Banco do Brasil's Big Data environment processing millions of daily transactions
 - Achieved **60% storage reduction** migrating legacy SQL tables to Parquet/Delta Lake
 - Reduced processing time by **50%** through Big Data cluster optimization at Claro Brasil
 - Cut maintenance time by **40%** via a reusable Python automation framework
 - Reduced manual intervention by **70%** through Azure Automation pipelines
 - Led **multi-cloud migration** (Oracle → AWS + Azure + GCP) of Cielo payment data
 - Delivered a healthcare Data Lake on **Apache Iceberg** with full **PII/LGPD governance** (Vault, KMS, Fernet)
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PROFESSIONAL EXPERIENCE

Data Engineer — BRQ

Jul 2025 – Present | Remote, Brazil

RD Saúde — Healthcare Data Lake (Customer 360)

- Built a healthcare Data Lake with Medallion architecture (Bronze/Silver/Gold) on **Apache Iceberg** over **S3**, serving analytical and downstream consumers via **Trino**, **Athena**, and Redshift, supporting electronic health records, scheduling, vaccines, medication, prescriptions, and tele-consultation use cases (Amplimed, HAOC, Vitat).
- Developed **PySpark/SparkSQL** pipelines for large-scale ingestion, transformation, and consolidation of clinical data, with support for **Apache Iceberg** and Apache Hudi over **Parquet**.
- Implemented an in-house **Data Quality** framework (cross-origin validations, light checks, DLS validations) and a YAML-versioned, self-service data catalog.
- Modeled the Customer 360 analytical table (dimensional modeling) integrating multiple sources (CRM, scheduling, events, interests), served to analytics and machine learning consumers.
- Orchestrated Airflow DAGs integrated with **Spark** on Kubernetes, with automated YAML manifest generation through Python builders.
- Integrated API + streaming-style ingestion (Amplimed, HAOC, Precifica) with async S3 clients and evolving schema handling (JSON/Parquet/Iceberg).
- Secured sensitive health data (PII/LGPD) with HashiCorp Vault, AWS KMS, and Fernet encryption, including decrypt pipelines in landing.
- Built end-to-end observability (custom log handlers, Spark metrics, exception tracking, Iceberg table comments) integrated with AWS SNS and Step Functions; maintained Docker containers (dev/qa/prod) for PySpark and Python K8s jobs.

Itaú — Modernization & Analytics

- Developed ETL pipelines using **EMR Serverless** and **Glue Studio**; optimized SQL queries for performance and implemented data quality processes.
 - Performed A/B test analysis and developed metric-improvement analyses and hypothesis validation.
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Senior Big Data Specialist / Data Engineer — Stefanini

Jun 2023 – Jun 2025 | São Paulo, Brazil

Banco do Brasil — Big Data Environment Sustainment

- Sustained and optimized a production Cloudera/Hortonworks cluster, ensuring **99.9% uptime SLA**.
- Developed Python and Shell automation for operational processes; built Jupyter notebooks for exploratory analysis and Ansible jobs for infrastructure management.
- Optimized Hive queries and maintained HDFS structures; built dbt models in Snowflake for financial reporting with data quality tests and versioned transformations.

AmericaNet — APIs & Data Processing

- Developed a RESTful API in Python Flask to expose processed data; implemented complex transformations with SparkSQL in Databricks.

- Automated processes with Azure Automation and built data pipelines with error handling and logging; built dbt/Snowflake ELT models with integrity validations and Unity Catalog.

Cielo — Dashboard Analytics

- Analyzed and documented legacy Oracle procedures; developed **PySpark** scripts for high-volume processing and **Athena** external tables over **S3** with partitioning.
- Migrated analytical data to Snowflake with dbt dimensional modeling supporting Power BI dashboards.

Itaú — ETL Process Refactoring

- Refactored legacy SQL tables to **Parquet** and converted stored procedures to **PySpark** jobs, achieving **60% storage reduction**.
- Built a data catalog with AWS **Glue**, implemented data quality processes, and produced detailed technical documentation of all migrated processes.

DirectLog — Geolocation System

- Built a pipeline capturing latitude/longitude via the Google Maps API and Airflow DAGs for batch address processing.
- Implemented an authenticated SFTP web interface, EC2 hosting, and a caching layer to optimize API calls.

Brasildev — AWS Data Lake

- Built a layered Data Lake (Bronze/Silver/Gold) with a data catalog and governance; created a Python ingestion engine with Salesforce integration.
- Configured security/access policies with Lake Formation and automated infrastructure with Terraform.

Senior Data Engineer — Natura & Co

Feb 2022 – Jun 2023 | São Paulo, Brazil

E-commerce Pipeline

- Modeled dimensional data for e-commerce analytics; developed ETL pipelines with **EMR Serverless** and Airflow DAGs for orchestration.
- Built Delta Lake structures for data versioning and incremental load processes; integrated with BI via Redshift.
- Built dbt models in Snowflake for customer-behavior analysis with data quality tests.

Cielo — Cloud Migration & Modernization

- Analyzed Oracle data structures and mapped them to BigQuery; migrated legacy data to Azure Blob Storage and built pipelines in Azure Data Factory with Unity Catalog cataloging.
- Converted complex SQL logic to distributed processing; developed **PySpark** and SparkSQL on Databricks; monitored spark-submit jobs.

JBS — Data Pipeline Modernization

- Provided technical leadership for a team of engineers and developers.
- Implemented a high-availability NiFi cluster integrated with Active Directory; configured an Application Load Balancer and built NiFi flows for SAP ingestion and S3↔Redshift movement (event-driven, streaming-style data movement).

Data Engineer — Enel

May 2021 – Feb 2022 | Brazil

Data Warehouse Migration

- Built a migration pipeline with Talend Open Studio moving data from Impala to Redshift.
- Created QlikSense dashboards connected to Redshift, documented existing ETL processes, optimized queries, and trained business teams on the new tooling.

Big Data Solutions Architect — Banco do Nordeste

Jan 2021 – May 2021 | Brazil

Corporate Data Lake

- Designed a Data Lake architecture with raw, stage, and analytics zones; implemented an unmanaged Hadoop cluster and configured Dremio for the analytics layer.
- Planned a NoSQL structure with Cassandra for a credit system; integrated legacy systems via DataStage and produced architecture documentation.

Bradesco — Data Ingestion Framework

- Updated ingestion-framework scripts for Hive and HBase tables; developed shell automation and integrated with the IBM IWS scheduler.
- Connected to Teradata for extraction and optimized load/transformation processes.

Big Data Architect / Engineer — Hospital Sírio-Libanês

Mar 2019 – Dec 2020 | São Paulo, Brazil

Hospital Analytics Platform

- Analyzed and ingested hospital data from multiple sources; installed and configured Airflow for orchestration and developed ETL DAGs.
- Implemented Docker containers integrated with Airflow, configured Kubernetes for scalability, integrated AWS Cognito with Azure AD for SSO, and set up a CloudWatch→Elasticsearch pipeline via **Lambda**.

Raízen — Data Ingestion Framework

- Surveyed the data-pipeline architecture and analyzed Azure Data Lake, Databricks, and Power BI; documented SAP integration via SSIS Theobald.

Dell EMC (Client: Fleury) — Consulting & Training

- Analyzed data and sustained a Hortonworks environment; configured security with Kerberos/Ranger and AD integration.
- Developed monitoring scripts for Oozie jobs, built Grafana monitoring dashboards, and delivered operational training to the client team.

Big Data Technical Lead — Claro Brasil

Sep 2017 – Mar 2019 | São Paulo, Brazil

Cloudera Cluster Implementation

- Installed and configured 3 Cloudera clusters for distinct projects (Data Lake, M2M, BI); implemented Kerberos with AD as KDC and cross-realm trust between clusters in different domains.
- Integrated Hive/Impala with BI tools (SAS, QlikView, Alteryx), built a data ingestion framework and monitoring/sustainment scripts, and integrated Informatica BDM for ETL — cutting processing time by **50%**. Mentored and led 2 engineers.

Nextel — Sustainment & ETL

- Developed ETL ingestion processes, analyzed data with Beeline/Impala, extracted data via Sqoop from relational sources, gathered functional requirements, and sustained the production environment.

Big Data Analyst — Vivo

Jun 2017 – Sep 2017 | Brazil

Hadoop Migration & Sustainment

- Built ETL ingestion processes, analyzed data with Hive and Python, migrated Hadoop 2.2 → 2.6, persisted data in HBase, and managed the cluster with Zookeeper.

OPEN-SOURCE PROJECTS ([GITHUB.COM/PAULOEDUSPBR](https://github.com/pauloeduspbr))

- **dune-analytics-queries** — 12 SQL queries on Dune analyzing **blockchain data**: DeFi protocol metrics, NFT markets, Ethereum and Solana activity, cross-chain comparisons.
- **crypto-data-pipeline** — production-style pipeline ingesting market data for 100 crypto assets (CoinGecko API) with Airflow, PostgreSQL, and Docker; 1.74s execution; 7/7 automated data quality checks passing.

- **medallion-data-lake** — production-ready Bronze/Silver/Gold template with Airflow, dbt, and Terraform (AWS + GCP), SCD Type 2 dimensions and 36 automated tests.
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EDUCATION

- **Associate Degree in Big Data and Data Intelligence** — Anhembí Morumbi University | Expected 2026
 - **Artificial Intelligence** (in progress) — FMU
 - **Bachelor of Science in Information Systems** — UNIP | 2006
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CERTIFICATIONS

- Red Hat Certified System Administrator (RHCSA)
 - Big Data Engineering — Garre Training (160h)
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LANGUAGES

- **Portuguese:** Native
- **English:** Upper-Intermediate (B2) — fluent technical reading and writing